

Lab 1

Experiment Name: IP address and Sub-netting.

Aim: Study of Network IP

- Introduction of IP Address.
- Classification of IP Address.
- Subnetting.
- Super Netting.
- Installation Cisco Packet Tracer.

Apperatus(Software): N/A Description:

Following is required to be study under this practical

Introduction of IP address:

“A unique numerical label assigned to a device connected to a network that uses the Internet Protocol for communication.”

While IP Address is associated with logical addressing and mac Address is associated with physical addressing. IPv4 is 32 bit while Mac Address is 48 bit. Ip address is divided into two types.

Based on Version:

- IPv4: A 32 bit address, the most commonly used IP Format
- IPv6: A 128 bit address, designed to replace IPv4 due to address Exhaustion.

Based on Accessibility:

- Public: A globally unique IP assigned by the Provider which is used as global identification for communication on internet.
- Private: A more locally Assigned IP to communicate within network without routing to internet.

Classification of IP Address: As shown in figure we get the idea regarding how the IP Address are classified and when there use cases.

Class	Address Range	Supports
Class A	1.0.0.0 to 126.255.255.254	Supports 16 million hosts on each of 127 networks
Class B	128.1.0.1 to 191.255.255.254	Supports 65,000 hosts on each of 16,000 networks.
Class C	192.0.1.1 to 223.255.254.254	Supports 254 hosts on each of 2 million networks.
Class D	224.0.0.0 to 239.255.255.255	Reserved for multicast groups.
Class E	240.0.0.0 to 254.255.255.254	Reserved.

Table 1

Note: as u might've noticed that 127.255.255.254 is missing it is because it is used for network loop back.

Subnetting:

The Process of Dividing a large IP Network into smaller, More Manageable sub- networks to improve efficiency, security and performance.

Class	Address Range	Default Mask	Subnet/CIDR Notation
Class A	1.0.0.0 – 126.255.255.255	255.0.0.0	/8
Class B	128.0.0.0 – 191.255.255.255	255.255.0.0	/16
Class C	192.0.0.0 – 223.255.255.255	255.255.255.0	/24

Table 2

Some Formulas:

Formula to calculate Sub-nets:

$$2^n$$

Formula to calculate Usable Hosts per subnet :

$$(2^h)-2$$

Example Regarding Subnetting:

For `192.168.1.0/26` (subnet mask `255.255.255.192`):

1. Subnets: ` $2^2 = 4$ ` (borrowed 2 bits from host portion).
2. Hosts per subnet: ` $(2^6)-2 = 62$ `.

Super netting:

The process to combine multiple smaller sub-nets into a larger network by modifying sub-net mask.

Installation of Cisco Packet Tracer:

1. Open Browser and go to <https://www.netacad.com/articles/news/download-cisco-packet-tracer> then press download button.

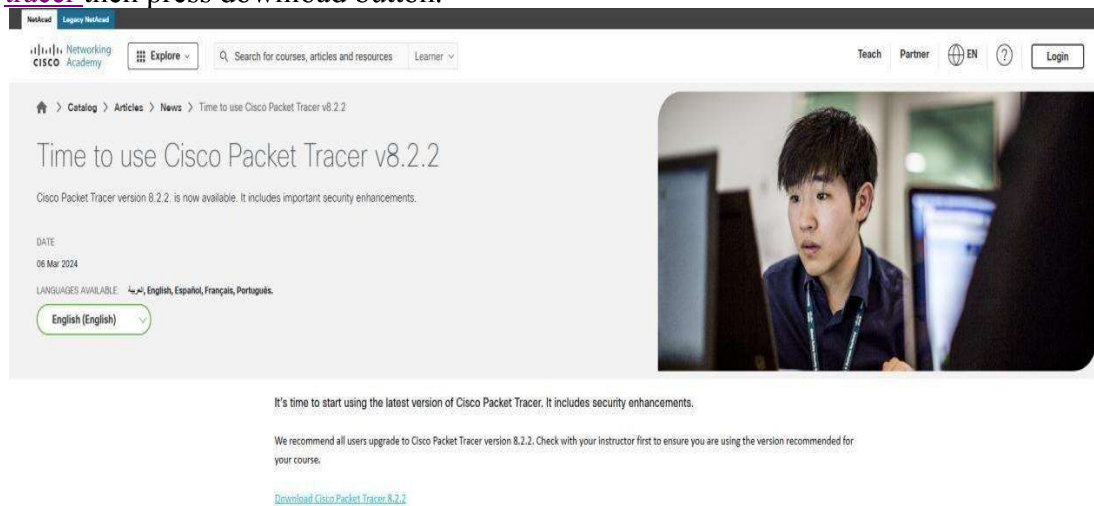


Figure 1

2. After that you will be redirected on the login page where you can use your university Email to register yourself.

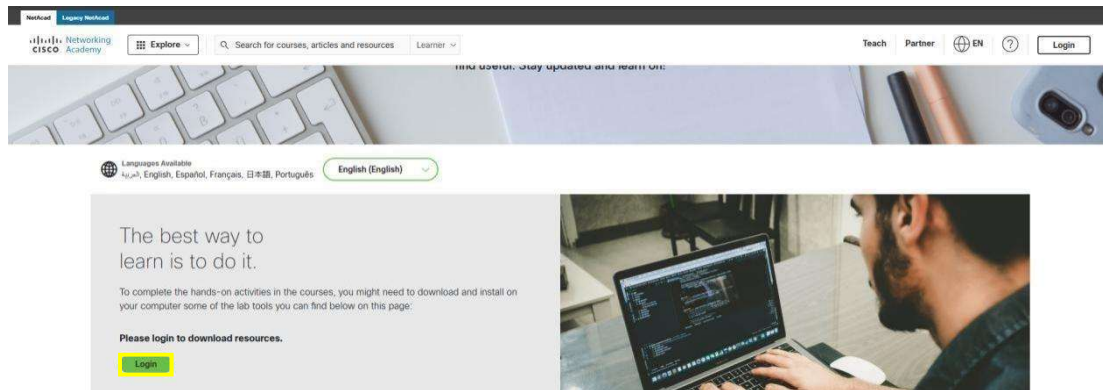


Figure 2

3. Once you register yourself you will be able to download the file.
4. After downloading the file just open the .exe file and follow the installation wizard.

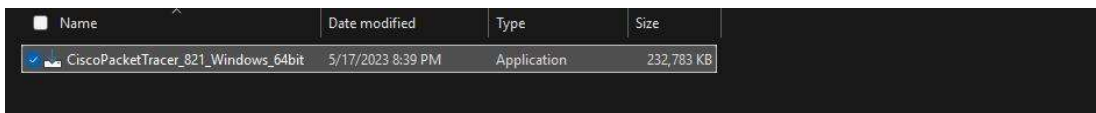


Figure 3

5. It will prompt you to Accept the Digital License Agreement, after accepting it just hit the install button.

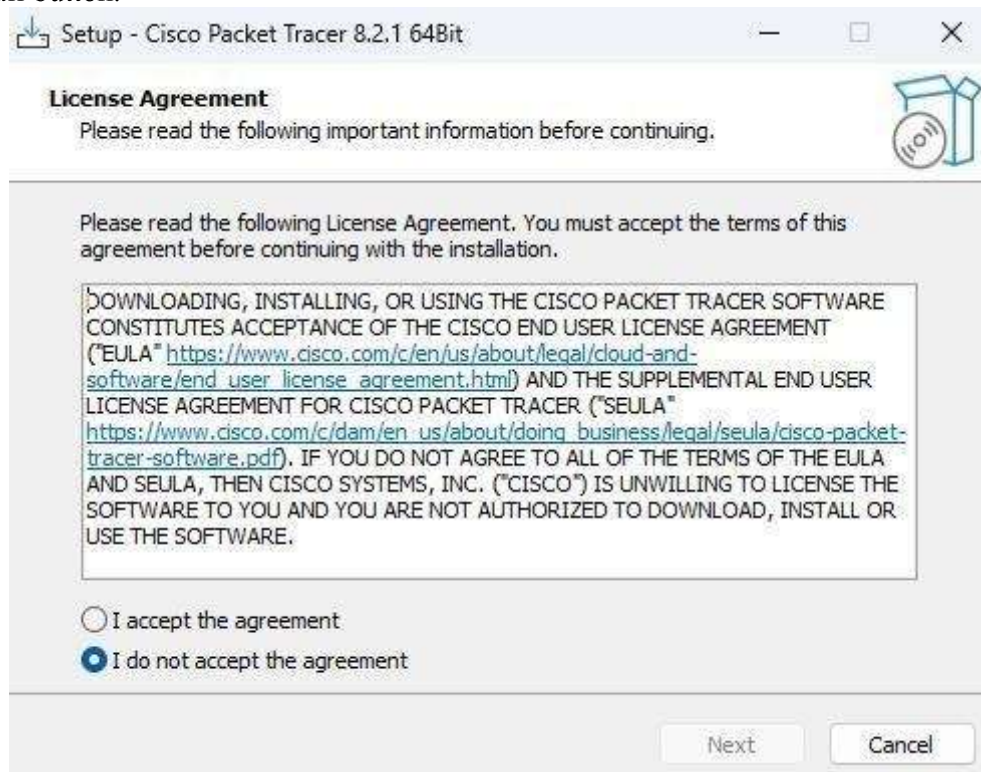


Figure 4

6. Once Installed, Click “Finish” to Complete.
7. After That Launch and login the account to start free trial (or Just Disconnect the network to so it doesn’t ask you to login).

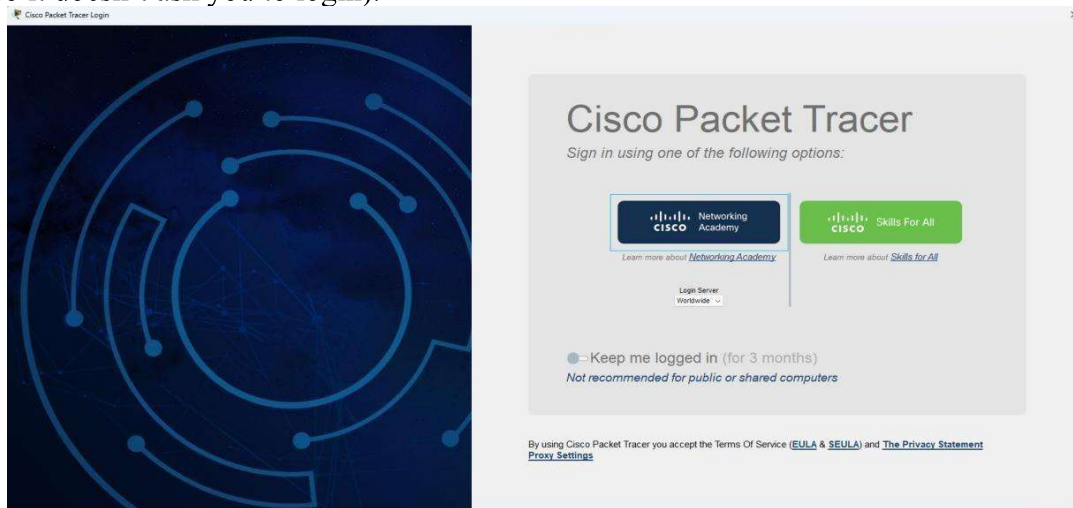


Figure 5